

The empirical recurrence for OEIS A254385: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A254385 records “Empirical recurrence of order 50”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 117$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 50, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 50 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A254385 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with every 3×3 subblock diagonal maximum plus antidiagonal maximum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

21304, 406548, 8533796, 175329586, 3587965682, 73923120984, 1521733260456, 31311650491630, ...

The entry states:

Empirical recurrence of order 50 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:38:19, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1748$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 117$ of the 1748, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 117$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 234$ exact terms of the model returns order 50 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{50} c_i a(n-i),$$

c_1	23	c_{14}	−23702032928	c_{27}	−3185462895788032	c_{40}	−629568218742128640
c_2	−22	c_{15}	−200481608192	c_{28}	39848377400631296	c_{41}	1258378020079534080
c_3	−604	c_{16}	709463535664	c_{29}	−79758251087298560	c_{42}	−642528485629231104
c_4	1553	c_{17}	−1811463143952	c_{30}	39788840039038976	c_{43}	16137740466782208
c_5	−21323	c_{18}	10925721868320	c_{31}	−3717469113942016	c_{44}	183910856097005568
c_6	374102	c_{19}	4403855164608	c_{32}	−48433316471898112	c_{45}	−429581649693376512
c_7	−7115148	c_{20}	−66760852800	c_{33}	113237205732294656	c_{46}	212847498174136320
c_8	9095944	c_{21}	−55937672341120	c_{34}	−3651899753234432	c_{47}	7063606294216704
c_9	146892908	c_{22}	−790036437982976	c_{35}	−115695910986448896	c_{48}	−10420071696433152
c_{10}	−148190520	c_{23}	1689699655347456	c_{36}	35586168391204864	c_{49}	20038599416217600
c_{11}	785046752	c_{24}	−1121483811830784	c_{37}	62900533836382208	c_{50}	−9618527719784448
c_{12}	−7134769952	c_{25}	−443185615263744	c_{38}	−85070442733240320		
c_{13}	8718510768	c_{26}	3059510887941120	c_{39}	69139754586734592		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 50, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $50 + 4$ terms removed returns order 50 as well, so $\deg q_{\min} = 50 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $50 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 50 that this sequence satisfies — and that family turns out to have exactly one member.

References

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